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MILITARY SPECIFICATION

ENGINES, AIRCRAFT, RECIPROCATING,
PRELIMINARY FLIGHT RATING TESTS FOR

This specification has been approved by the Department of the Air Force and by the Navy Bureau of Aeronautics.

1. SCOPE

1.1 This specification covers the Preliminary flight rating test requirements for reciprocating aircraft engines.

2. APPLICABLE DOCUMENTS

2.1 The applicable publications listed in the following bulletin, of the issue of the bulletin in effect on date of invitation for proposals, form a part of this specification:

PUBLICATIONS

Air Force-Navy Aeronautical Bulletin

No. 343 Specifications and Standards Applicable to Aircraft Engines and Propellers, Use of

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Approval of preliminary flight rating.-

3.1.1 Approval of the preliminary flight rating shall be predicated on satisfactory completion of the tests specified herein.

3.1.2 The engine shall be considered to have satisfactorily completed the 50-Hour endurance test if it is operating satisfactorily at the end of the test and if the recalibrations and teardown inspection do not reveal any indication of imminent failure or performance change which would compromise safety of flight. In lieu of a repetition of the test, in the event there is indication of imminent part failure which would compromise safety of flight, the Government may, at its option, authorize the contractor to reassemble the engine without parts change and conduct an additional 10-hour endurance run to the specified schedule to prove the part in question.

3.2 Reports - engine and components.-

3.2.1 Reports.- Immediately following completion of the engine tests and each separate component test, or consecutive group of tests conducted on any single test

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assembly or components, a report shall be submitted. This report, combined with the certification of a Government representative as to proper conduct of the tests, will normally be used as a basis for approval of the tests.

3.2.1.1 Preparation.- This report shall contain essentially the following information:

(a) General summary of test, giving dates, failures, test incidents, performance changes, marginal conditions, etc.

(b) Description of the condition of the engine and components at disassembly inspection.

(c) Recommendations, with respect to approval of the engine and components, supplemented by such discussion as is necessary for their justification.

(d) Calibration and recalibration data: (Data in both uncorrected and corrected form, if applicable, shall be shown by suitable curves.)

(e) Tabulated data (when requested by the using service) (tabulated data of all instrument readings taken during the test).

3.2.1.2 Number and distribution of copies.- When the engine Preliminary flight rating tests are conducted at the contractor's plant, five copies of the report thereon shall be forwarded to the Commander, Wright Air Development Center, Wright-Patterson Air Force Base, Ohio, Attention: Power Plant Laboratory; and three copies shall be forwarded to the Chief, Bureau of Aeronautics, Navy Department, Washington 25, D. C., Attention: Power Plant Division.

4. QUALITY ASSURANCE PROVISIONS

4.1 Accuracy of data.- For all engine and component tests, the test apparatus shall be such as to insure that reported data shall have a static accuracy within 2 percent of the values obtained at the takeoff rating of the engine, except for fuel flow which shall be accurate within 1 percent, and rpm which shall be accurate within 0.2 percent of the values obtained at the takeoff rating of the engine. For engine and component calibrations, the test apparatus shall provide the same degree of accuracy with the added requirement that shaft horsepower shall be accurate within 1 percent of the values obtained at the takeoff rating of the engine. All instruments and equipment shall be calibrated often enough to insure that this degree of accuracy is maintained.

4.2 Engine tests.-

4.2.1 Test conditions.-

4.2.1.1 Test apparatus.-

4.2.1.1.1 Power absorption.- Calibration shall be conducted on a type of power absorbing dynamometer acceptable to the using service, such as an electric cradle dynamometer, or water brake unless the endurance test is to be conducted with an approved calibrated torque measuring device. The endurance test may be conducted on either a reaction or rigid test stand. Specific authorization of the using service is required to conduct the endurance test on a dynamometer or other power absorbing device which does not transmit propeller thrust to the engine propeller shaft. The type and model of propeller shall be acceptable to the using service.

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4.2.1.1.1.1 Torsional vibration.- Torsional vibration tests shall be made on the dynamometer before the Power calibration, and speeds corresponding to resonance periods shall be subsequently avoided as far as practicable. When the Preliminary flight rating tests are conducted by the contractor, the Torsional vibration test on a dynamometer may be omitted.

4.2.1.1.2 Sludge filter.- Sludge removing filters, not supplied with the engine, shall not be used.

4.2.1.1.3 Revolutions per minute.- The crankshaft revolutions per minute shall be determined by means of a positive counter which actually will count the revolutions for a period of not less than 1 minute. A combination unit which counts the revolutions while measuring the time required to consume the specified weight or volume of fuel, is preferred.

4.2.1.1.4 Fuel flow.- Fuel flow measurement shall be made by either the volume or weight method. The quantity selected for the volume or weight method shall be such that each reading will cover an elapsed time of at least 1 minute. Flowmeter readings for calculations of specific fuel consumption shall be used only when the flowmeter, and the use thereof, has been approved by the procuring activity. Regardless of the method used in determining fuel and oil consumption, all expressions of quantities shall be recorded on a weight basis.

4.2.1.1.5 Airflow.- For all engines, airflow measurements shall be made by a method acceptable to the using service, preferably by use of a smooth-approach nozzle or venturi tube.

4.2.1.1.6 Water vapor pressure determination.- Vapor pressure shall be determined by means of a vapor pressure chart in accordance with Drawing AND10229 from readings of wet and dry bulb thermometers, or other method acceptable to the using service. When using atmospheric air, wet and dry bulb thermometers may be located in such a position as to insure measurement of humidity of air entering the engine with reasonable accuracy. When using conditioned air, the location of thermometers for determining vapor pressure shall be acceptable to the using service. Air velocity over the wet bulb thermometer shall be sufficient to insure reasonable accuracy. The thermometers shall be so arranged that the air passing over the wet bulb will not subsequently pass over the dry bulb. The wick of the wet bulb shall be kept wet with distilled water only. The wick of the wet bulb shall be replaced whenever a perceptible quantity of oil, dirt, incrustation, or other foreign substances, has accumulated thereon.

4.2.1.1.7 Temperature measurements.- Cylinder head temperatures of air-cooled engines shall be measured at the position specified on the installation drawing. The cylinder barrel temperature shall be taken on the downstream side of the cylinder in the fillet where the cylinder barrel joins with the cylinder holddown flange. In the case of liquid-cooled engines the cylinder temperature shall be taken at the hottest spark-plug gasket. All temperatures shall be reported in degrees Fahrenheit. Other thermocouple locations may be used in lieu of those mentioned above where called for in the model specification.

4.2.1.2 Test engines.- Two identical engines may be used for the tests required by this specification. One of the engines shall be used for calibration and the other engine shall be used for the 50-Hour endurance run and all other specified tests, provided, however, that the tests specified for "Power calibration" 4.2.2.1 may, at the option of the using service, be accomplished on either engine. If only one engine is used for the tests required by this specification, the altitude calibration and other similar Special tests, if required, shall be accomplished only after completion of the 50-Hour endurance run, the Dive tests, and the subsequent Teardown inspection.

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4.2.1.3 Preliminary data.- The engine weight, if not previously obtained, photographs, and other pertinent engine data desired by the using service shall be obtained preferably at the time the engine is being prepared for test. At the option of the using service, the engine may be disassembled for examination or measurement of parts prior to testing.

4.2.1.4 Operating test conditions.-

4.2.1.4.1 General.- For all testing, the fuel octane or performance number rating, grade of oil, induction system cooler outlet temperatures, and coolant outlet temperature shall be as specified in the model specification. For air-cooled engines, for all operating periods longer than 10-minute duration, the temperature of the hottest cylinder head shall be that specified for cylinder head temperatures in the model specification, within a tolerance of plus zero minus 30 Fahrenheit degrees, provided that in no case shall the cylinder barrel temperatures exceed those specified in the model specification. When cylinder barrel temperatures become the limiting factor, they shall be held to a tolerance of plus zero minus 15 Fahrenheit degrees. For liquid-cooled engines, the coolant outlet temperature shall be maintained at the values specified in the model specification, within a tolerance of plus or minus 5 Fahrenheit degrees. Temperature for dives and idle runs shall be as indicated by test conditions. During takeoff and military power runs, the minimum allowed temperature shall be reached with the first 4 minutes of the run.

4.2.1.4.2 Test fuel.- Fuel containing the maximum tetraethyl lead content permitted by Specification MIL-F-5572 shall be used for all testing.

4.2.1.4.2.1 Fuel knock rating.- The knock rating of the fuels specified for use in testing shall not exceed the minimum knock rating by more than three performance numbers for fuels having normal rating of 100 octane, or higher, nor more than two octane numbers for fuel having normal rating below 100 octane.

4.2.1.4.3 Test oil.- The test oil shall not be required to meet any requirements in addition to those listed in Specification MIL-L-6082.

4.2.1.4.4 Supercharger clutch shifts.- No supercharger clutch shifts shall be made, except those required in the test schedule.

4.2.1.4.5 Oil inlet temperatures.- The oil inlet temperature with oil conforming to Specification MIL-L-6082, grade 1100, shall be maintained at 185°F, except for takeoff and military rated power runs when the oil inlet temperature shall be maintained at 203° ±5°F. In engines in which the grade of oil or oil inlet temperature as specified in the model specification are different from the above, the grade of oil and the oil inlet temperature as specified in the paragraph entitled "Oil flow and heat rejection" of the engine model specification shall be used with ±5°F temperature limits.

4.2.1.4.6 Oil pressures.-

4.2.1.4.6.1 Main oil pressure(s).- Oil pressure(s) shall be adjusted at normal rated power and speed to the value specified in the model specification, and the only adjustments in the relief valve(s) permissible shall be those necessary to maintain the specified pressure at normal rated speed.

4.2.1.4.6.2 Oil outlet pressure.- The oil outlet pressure shall be maintained at 40 pounds per square inch minimum during the testing specified herein, except for tests at less than 60-percent normal rated speed when it may be less, as indicated by test conditions.

4.2.1.4.7 Carburetor air temperatures.- During all tests, except those under altitude conditions, the air temperature at the carburetor or air throttle entrance shall be maintained between 75°F and 90°F, except for engines rated for use with multistage or turbosupercharger, the temperature shall be as specified in the engine model specification within plus 10 Fahrenheit degrees and minus zero Fahrenheit degrees but not less than 100°F. During tests under altitude conditions with single-stage geared supercharger

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engines, the air temperature at the carburetor or air throttle entrance shall be maintained at the standard altitude value as closely as practicable.

4.2.1.4.8 Carburetor air and exhaust pressure.- During portions of the test in which the engine power corresponds to a rating for which external or multistage supercharging is required at sea level, the carburetor or air throttle entrance pressure and the exhaust pressure shall be maintained at values simulating operation at the particular power with the manner of supercharging to be employed. For engines intended for use with a turbine supercharger, but for which external supercharging is required to obtain takeoff and military ratings at sea level, an absolute pressure of 32 to 33 inches of mercury shall be applied to the exhaust during all endurance operations at takeoff and military power.

4.2.1.4.9 Engine speed.- At takeoff and military speed, the engine speed shall be controlled to within ±2 percent of the stipulated speed. At normal rated speed or below, the engine speed shall be controlled to within ±3 percent of the stipulated speed. Except where specified for supercharger gear shift runs, no portion of the 50-Hour endurance run shall be accomplished with the propeller pitch controlled by a constant speed governor, unless the engine is equipped with a torquemeter, or unless specific authorization is obtained from the using service.

4.2.1.4.10 Power.- Actual brake horsepower and not corrected brake horsepower shall be used for all endurance tests, and the absolute manifold pressure shall be maintained to within plus 1-1/2 percent and minus zero percent of the value required to develop the specified power. When the engine is delivered with a torquemeter, a torquemeter shall be used and the actual brake horsepower shall be considered as equal to the sum of the indicated torquemeter horsepower and the power absorbed by the accessories.

4.2.1.4.11 Fuel and oil consumption.- The specific fuel consumption during each period of the 50-Hour endurance run shall be maintained within the limits of plus 3 percent and minus zero percent of the values specified in the model specification for the corresponding powers and speeds. All periods between the maximum and minimum powers stipulated in the model specification for operation at best economy shall be run with best economy mixture strength; but in no case shall the mixture strength be leaner than the leanest mixture strength permitting smooth operation. For runs of altitude engines at normal rated power at other than standard normal rated altitude conditions, the values of fuel consumption shall be those specified for the normal rated altitude. The specific oil consumption shall not exceed the value specified in the model specification.

4.2.1.4.12 Accessory drives.- Unless otherwise authorized by the using service, actual accessories or suitable loading devices, the type of each to be acceptable to the using service, shall be installed on all accessory drives and operated at normal loads during the 50-Hour endurance run. Power calibrations of the engine shall be made without loading accessory drives.

4.2.1.4.12.1 Power takeoff drive loading.- When required by the using service, the power takeoff drive shall be loaded by a device adjusted to impose, when the engine is operated at normal rated speed, a torque within plus 5 percent and minus zero percent of the continuous torque rating specified for the drive in the specification under which the engine was built. At engine speeds higher than normal rated speed, the torque imposed shall equal the specified torque rating within the limits of plus 10 percent, minus zero percent. At engine speeds lower than normal rated speed, the torque imposed shall equal the specified torque rating within the limits of plus 5 percent, minus 15 percent.

4.2.1.4.12.2 Propeller governor pressure loading.- When a propeller governor is installed, it shall be adjusted to impose pressure not exceeding that specified in Specification MIL-E-25109 upon any passages in the engine for conveying fluid to the propeller, during such portions of the test as the using service may direct.

4.2.1.4.12.3 Starter.- A starter, of a type agreed upon by the contractor and the using service, shall be used to make all engine starts, when the endurance testing is accomplished with a propeller.

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4.2.1.4.13 Inspections.- Unless otherwise specified by the using service, or otherwise recommended by the contractor and authorized by the using service, no inspections of the engine shall be made other than external visual inspections during the 50-Hour endurance run. Disassembly of the engine to any extent prior to the final teardown inspection shall be at the option of the using service.

4.2.1.4.14 Oil servicing.- The oil system shall be filled with new oil at the start of the endurance test and thereafter makeup oil shall be added as required. The oil system shall not be drained during the test.

4.2.1.4.15 Accreditable test time.- Intervals of endurance test operation of less than 1 hour's duration terminated by an engine failure shall not be credited to the required test time. Endurance test time shall not be credited by increments shorter than 30 minutes, except when shorter periods are a test requirement.

4.2.1.4.16 Fuel and oil consumption calculations.- Specific fuel and oil consumption during the 50-Hour endurance run shall be calculated on the basis of actual horsepower output, as defined in 4.2.1.4.10 entitled "Power."

4.2.1.4.17 Water vapor pressure data.- Wet and dry bulb air temperature readings shall be taken at intervals not exceeding 3 hours.

4.2.1.4.18 Barometer readings.- The room barometer shall be read and recorded at intervals not exceeding 3 hours.

4.2.1.4.19 Corrections.-

4.2.1.4.19.1 Horsepower calculations.- To compensate for the variations in the temperature of the air entering the engine, the manifold pressure required to develop the actual power output specified shall be calculated on the assumption that with constant manifold pressure the actual horsepower is inversely proportional to the square root of the absolute carburetor or air throttle entrance temperature. An alternative temperature correction may be used if based on substantiating test data acceptable to the using service.

4.2.1.4.19.2 Manifold pressure test values.- All test values shall be on a basis of dry absolute manifold pressure which is obtained by subtracting the water vapor pressure from the observed absolute manifold pressure.

4.2.1.4.19.3 Correction of the barometer or mercury manometer for temperature.- The observed reading of a mercury column varies not only with changes in the applied pressure, but also with changes in the temperature of the mercury and of the scale. The observed readings must, therefore, be corrected to the values they would have if the mercury and scale were maintained at their standard temperatures. The correction given in table I, corresponding to the observed height of the mercury column and to the temperature indicated by a thermometer attached to the column, is to be added to or subtracted from the observed height to obtain the true height or, in the case of a barometer, the true barometric pressure. This correction shall be made for all readings greater than 10 inches. Only the true barometer reading shall be recorded on the log sheet.

4.2.2 Methods of test.-

4.2.2.1 Power calibration.- Power calibration runs shall be made in accordance with the requirements of the using service. Data obtained shall be used to adjust the throttle and to obtain the required power during the 50-Hour endurance run. Operating time on the dynamometer, before the 50-Hour endurance run, shall be limited to the minimum practicable.

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TABLE I

Temperature corrections for mercury columns 1/

Temp of column °F	Observed reading of column in inches of mercury													
	10	15	20	25	30	35	40	45	50	55	60	65	70	75
	Add													
-20	.05	.07	.09	.12	.14	.16	.19	.21	.24	.26	.28	.31	.33	.35
-10	.04	.06	.08	.10	.11	.13	.15	.17	.19	.21	.23	.25	.27	.29
0	.03	.04	.06	.07	.09	.10	.12	.13	.14	.16	.17	.19	.20	.22
10	.02	.03	.04	.05	.06	.07	.08	.09	.10	.11	.12	.13	.14	.15
20	.01	.02	.02	.03	.03	.04	.04	.05	.05	.06	.07	.07	.08	.09
30	0	0	0	0	.01	.01	.01	.01	.01	.01	.01	.01	.01	.01
	Subtract													
35	0	0	.01	.01	.01	.01	.01	.01	.01	.01	.02	.02	.02	.02
40	.01	.01	.01	.01	.02	.03	.03	.03	.04	.04	.04	.05	.05	.05
45	.01	.02	.02	.03	.04	.04	.05	.05	.06	.06	.07	.08	.08	.09
50	.02	.02	.03	.04	.05	.06	.06	.07	.08	.09	.10	.11	.11	.12
55	.02	.03	.04	.05	.06	.07	.08	.09	.10	.11	.13	.14	.15	.16
60	.03	.04	.05	.06	.08	.09	.10	.11	.13	.14	.15	.16	.18	.19
65	.03	.04	.06	.07	.09	.10	.12	.13	.15	.16	.18	.19	.21	.22
70	.03	.05	.07	.09	.10	.12	.14	.15	.17	.19	.21	.22	.24	.26
75	.04	.06	.08	.10	.11	.14	.16	.18	.19	.21	.23	.25	.27	.29
80	.04	.07	.09	.11	.12	.15	.17	.20	.22	.24	.26	.28	.30	.33
85	.05	.07	.10	.12	.13	.17	.19	.22	.24	.26	.29	.31	.34	.36
90	.05	.08	.11	.13	.16	.18	.21	.24	.26	.29	.31	.34	.37	.39
95	.06	.09	.11	.14	.17	.20	.23	.26	.29	.31	.34	.37	.40	.43
100	.06	.09	.12	.15	.18	.22	.25	.28	.31	.34	.37	.40	.43	.46

1/ The table gives corrections in inches for the observed heights in inches and the temperatures of the mercury column in degrees Fahrenheit. The table is computed for columns measured with a brass scale but is applicable with the negligible error to measurements with scales of other materials.

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4.2.2.2 50-Hour endurance run.- Following the Power calibration, the engine shall be given the following 50-Hour endurance run, conducted, unless otherwise specified by the using service, in the order given:

(a) First period: 10 hours of alternate periods of 2-1/2 hours each at normal rated power and speed, and at 90-percent normal rated power and 97-percent normal rated speed.

(b) Second period: 6 hours of alternate periods of 5 minutes at takeoff power and speed, and 10 minutes at as low an idling speed as practicable.

(c) Third period: 6 hours of alternate periods of 30 minutes at military rated power and speed, and 30 minutes at a selected speed between 55 and 65 percent of military rated speed without change of propeller pitch. In the absence of a military rating, 6 hours of the test for the second period shall be repeated in lieu of this requirement.

(d) Fourth period: 10 hours of alternate periods of 2-1/2 hours each at normal rated power and speed, and at the maximum best economy cruise power as stipulated in the model specification.

(e) Fifth period: 10 hours at alternate periods of 2-1/2 hours each at 80-percent normal rated power and 93-percent normal rated speed, and at 80-percent normal rated power and 89-percent normal rated speed.

(f) Sixth period: 6 hours at 42-percent normal rated power and 56-percent normal rated speed, or at the speed necessary to obtain 42-percent normal rated power with full throttle. For engines not intended for use with controllable speed propellers, the speed shall be 75-percent normal rated speed.

(g) Seventh period: 2 hours at normal rated power and 110-percent normal rated speed.

4.2.2.3 Special tests.-

4.2.2.3.1 Single-stage, two-speed, supercharged engines.- For engines equipped with two-speed superchargers, the following running, conducted on a test stand with simulated altitude conditions shall be substituted for that portion of the 50-Hour endurance run, specified in 4.2.2.2 and 4.2.2.3.3, as applicable. (The simulated altitude test equipment shall be acceptable to the using service.) Where simulated altitude test equipment is not available and when operating in the higher gear ratio, it shall be permissible to run at the observed horsepowers obtained with the normal or military rated manifold pressure for the higher gear ratio, or the specified percentages thereof. When run in this manner it shall also be permissible to run at fuel-air mixture strength rich enough to suppress detonation.

(a) Substitution running period: The following running period shall be substituted for the "First period," 4.2.2.2(a), and for the "Second period," 4.2.2.3.3(b):

(1) 2 hours at normal rated power and speed at the lower rated altitude in the lower gear ratio.

(2) 2 hours at normal rated power and speed at the higher rated altitude in the higher gear ratio.

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(3) 2 hours at 90-percent normal rated power and 97-percent normal rated speed at the higher rated altitude in the higher gear ratio.

(4) 4 hours of alternate periods of 5 minutes in each of the supercharger gear ratios. The first 5 minutes of the test shall be conducted at normal rated speed with the higher gear ratio and the observed horsepower obtainable with 90 percent of normal rated manifold pressure for the higher gear ratio under sea level conditions. The conditions for operation for the alternate 5 minutes in the lower gear ratio shall be that obtained by simply shifting to the lower gear ratio. This test shall be conducted under normal sea level conditions and with a constant speed propeller, unless otherwise specified by the using service.

(b) Substitution running period: For engines with a military rating in the higher gear ratio, 2 hours of the running specified for the "Third period," 4.2.2.2(c), and for the "First period," 4.2.2.3.3(a), shall be conducted on a simulated altitude test stand in the higher gear ratio.

(c) Remainder of endurance run.- Engines equipped with two speed superchargers shall be operated in the lower gear ratio at powers based on the normal rated power at the lower normal rated altitude, except when the use of the higher gear ratio is required for operation at takeoff or military rated power and further, except that during the last 1/2 hour of the running specified for the "Seventh period," 4.2.2.2(g) and the "Seventh period," 4.2.2.3.3(g), the engine shall be operated for two 10-minute periods spaced 10 minutes apart in the higher gear ratio at not less than 60-percent high gear ratio normal rated power nor less than 70-percent high gear ratio normal rated speed.

4.2.2.3.2 Engines with special characteristics.- For engines having unusual vibration characteristics or special features or ratings, appropriate tests may, by mutual agreement between the using service and the contractor, be substituted for portions of the 50-Hour endurance run, 4.2.2.2 or 4.2.2.3.3. This test time will be deducted from the appropriate items.

4.2.2.3.3 50-Hour endurance run, helicopter engines.- The following schedule of running in lieu of the 50-Hour endurance run specified in 4.2.2.2 shall be conducted on helicopter engines:

(a) First period: 10 hours of alternate periods of 30 minutes at military rated power and speed and 30 minutes at normal rated power and speed.

(b) Second period: 10 hours continuous operation at normal rated power and speed.

(c) Third period: 8 hours continuous operation at 90-percent normal rated power and 100-percent normal rated speed.

(d) Fourth period: 6 hours continuous operation at 80-percent normal rated power and military rated speed.

(e) Fifth period: 10 hours of alternate periods of 2-1/2 hours at 80-percent normal rated power and 100-percent normal rated speed and 2-1/2 hours at 70-percent normal rated power and 96-percent normal rated speed.

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(f) Sixth period: 3 hours continuous operation at 105-percent normal rated power and 100-percent normal rated speed.

(g) Seventh period: 3 hours continuous operation at 100-percent normal rated power and 110-percent normal rated speed.

NOTE: If the engine does not have a military rating, the "First period" shall consist of 5 hours of running at normal rated power and speed and 5 hours at normal rated power and 110 percent normal rated speed.

4.2.2.4 Miscellaneous tests.- The following tests shall, except as otherwise specified, be conducted after the 50-Hour endurance run or, alternatively, at any convenient time during the interval in which the 50-Hour endurance run is made.

4.2.2.4.1 Ignition system check.- During the 50-Hour endurance run, observations shall be made with an ignition analyzer of any nonfiring cylinders as well as general engine operation. Abnormal patterns shall be investigated at the end of the 50-Hour endurance run.

4.2.2.4.2 Radio interference level.- The radio interference level of all electrical and electronic equipment associated with the engine shall be tested in accordance with Specification AN-I-27 or MIL-I-6181, as applicable, except as specified in the paragraph entitled "Electrical interference" of Specification MIL-E-25109.

4.2.2.4.3 Lubrication check.- Observations of engine oil pressures shall be made to determine conformance with applicable requirements of the contract and specifications under which the engine was built.

4.2.2.4.4 Dive tests.- Upon completion of the 50-Hour endurance run, Special tests, and miscellaneous tests, and without intervening major disassembly, the engine shall be subjected to 10 dives of 30 ±3 seconds duration each at 120-percent normal rated speed. Dive tests on multispeed- or multistage-supercharged engines shall be accomplished with the lowest degree of supercharging only.

4.2.2.4.4.1 Dive test procedure.- Unless otherwise specified by the using service, the Dive tests shall be conducted as follows:

(a) All dives shall be made with a propeller.

(b) Dives shall be conducted preferably at the lowest manifold pressure which will give satisfactory firing of the engine.

(c) Dives shall be alternated with stabilizing runs of about 5 minutes duration each, at from 60 to 80 percent of normal rated speed. Acceleration and deceleration shall each be accomplished in a period not longer than 10 seconds. The time for changing speed shall not be deducted from the specified duration time for dives.

4.2.2.4.5 Uncoupled run, helicopter engines.- Uncoupled runs shall be conducted in lieu of the Dive tests of 4.2.2.4.4 on helicopter engines. Upon completion of the 50-Hour endurance run and without intervening major disassembly, the engine shall be subjected to 50-uncoupled test runs, each consisting of a start, 5-minute run at the lowest practical idling speed, and a run of 30 ±3 seconds duration at 120-percent normal rated speed after which the engine shall be stopped.

4.2.2.4.5.1 Uncoupled run, test procedure.- Unless otherwise specified by the using service, the uncoupled run shall be conducted as follows:

(a) For all uncoupled runs, the engine shall be loaded only with a suitable helicopter fan and flywheel assembly, or with a flywheel of the minimum polar moment of inertia as specified in the model

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specification. Acceleration and deceleration from the idling speed to 120-percent normal rated speed shall be accomplished in the shortest practical time. The time for throttle lever movement shall not exceed 1 second. The time for changing speed shall not be deducted from the specified time at 120-percent normal rated speed.

4.2.2.4.6 Altitude test, ignition system.- Altitude performance of the ignition system shall be demonstrated in a pressure chamber acceptable to the using service and capable of containing the ignition system. Means of maintaining required pressure and of determining simulated density altitude shall be provided. With the ignition system installed in the chamber, and operating at the equivalent of the highest rated engine speed, the simulated density altitude shall be increased to the maximum prescribed and maintained for 6 hours without failure. During this test the voltage dissipated by the system must be maintained equivalent to the maximum engine requirement.

4.2.2.5 Power calibration check.- If the calibration prior to the endurance run and the endurance run are accomplished by separate agencies, or if malfunctioning of the engine during the endurance run indicates the desirability of a recalibration, a calibration check run during which engine-running time at or more than 75 percent of normal rated power shall not exceed 1 hour, shall be made at the option of the using service, after the completion of the 50-Hour endurance run and Miscellaneous tests. The indicated decrease in power at normal rated speed at equivalent operating conditions from the previous calibration shall not exceed 5 percent.

4.2.2.6 Data.-

4.2.2.6.1 50-Hour endurance run.- During the 50-Hour endurance run, except for the testing covered by the second and third periods, at least the following data, where applicable, shall be recorded at intervals not greater than 30 minutes.

(a) Time of day.

(b) Total endurance time.

(c) Revolutions per minute of crankshaft.

(d) Torque (if obtained).

(e) Absolute manifold pressure.

(f) Intake manifold temperature.

(g) Absolute air pressure at inlet to engine.

(h) Air temperature at inlet to engine.

(i) Crankcase pressure.

(j) Cooling air temperature.

(k) Liquid coolant temperature at inlet to pump.

(m) Liquid coolant temperature leaving engine.

(n) Liquid coolant flow.

(o) Liquid coolant pressure at inlet to pump.

(p) Liquid coolant pressure leaving engine.

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(q) Oil temperature at inlet to pump.

(r) Oil temperature leaving engine.

(s) Oil pressure (engine).

(t) Oil pressure at inlet to pump.

(u) Oil pressure at outlet of engine.

(v) Oil flow.

(w) Oil scale reading.

(x) Fuel consumption.

(y) Fuel pressure.

(z) Supercharger gear ratio.

(aa) Airflow.

(bb) If air-cooled, the cylinder head and base temperatures of all cylinders.

(cc) If liquid-cooled, the temperatures at at least two cylinders taken at the hottest spark plug gasket.

4.2.2.6.1.1 During the testing covered by the second and third periods, not more than the initial and final 15 seconds of each 10-minute or 30-minute period at the lower speed shall be used for deceleration and acceleration. The following readings, where applicable, shall be taken once during each period:

(a) Total endurance time.

(b) Revolutions per minute of the crankshaft.

(c) Absolute manifold pressure.

(d) Intake manifold temperature.

(e) Air temperature at entrance to engine.

(f) Liquid coolant temperature leaving engine.

(g) Coolant flow.

(h) Oil temperature at inlet to pump.

(i) Oil temperature leaving engine.

(j) Oil pressure in pounds per square inch.

(k) Fuel pressure in pounds per square inch.

(m) Fuel consumption.

(n) Crankcase pressure.

(o) If air-cooled, the cylinder head temperature and the base temperature on the cylinder which has the highest temperature.

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4.2.2.6.2 Miscellaneous tests.- For each dive and each stabilizing run or uncoupled run, at least the following data shall be recorded except as noted:

(a) Tachometer rpm.

(b) Absolute manifold pressure.

(c) Oil pressure (engine).

(d) Oil temperature at inlet to pump.

(e) Oil temperature leaving engine.

(f) Oil flow, when measurement is practicable.

(g) Oil pressure at outlet of engine (stabilizing run only).

(h) Crankcase pressure.

4.2.2.6.3 Log sheet.- The data, desired power and speed, model of spark plugs, supercharger gear ratio in use, type of air heater or hotspot and whether in use or not, the engine model designation, and the serial number of the engine shall be recorded on each log sheet.

4.2.2.6.4 Incidents.- Notes shall be placed on the log sheets of all incidents of the run, such as change of spark plugs, valve clearance adjustments, leaks, vibration, and any other irregular functioning of the engine or the equipment and corrective measures taken.

4.3 Teardown inspection.- After completion of the above tests, the engine shall be completely disassembled for examination of all parts and measurement as necessary to disclose excessively worn, distorted, or weakened parts. These measurements shall be compared with the contractor's drawing dimensions and tolerances, or with similar measurements made prior to the test, when available.

4.4 General inspection.- All tests shall be subject to witnessing by a Government representative. At convenient times prior to the tests and during Teardown inspection, the engine and components shall be examined to determine if they conform to all requirements of the contract and specifications under which they were built. At no time during the test shall any part of the engine be removed, disassembled, or adjusted for examination, without prior approval of the Government representative.

4.5 Engine parts' failures.- If, during the Preliminary flight rating tests a part fails, this part may be replaced, or a new test started on a new engine or component with a redesigned part or one of different material, unless the using service authorizes the installation of a new part of original design and material for one which, in the judgment of the using service, failed owing to faulty material or workmanship. The Preliminary flight rating tests shall be considered complete when in the judgment of the using service every part has satisfactorily completed the tests.

5. PREPARATION FOR DELIVERY

5.1 Not applicable to this specification.

6. NOTES

6.1 Intended use.- This specification covers the test requirements for reciprocating aircraft engines which will have a preliminary flight rating established with

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the objective of insuring that the engines have a sufficient durability and reliability to permit experimental flight test with a minimum of 50 hours of operation before overhaul.

6.2 Definitions.- Definitions of the terms used herein will be as specified in Specification MIL-E-25110.

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